GeoPIXE Recent News

Version 8.9n

FalconX (Merge) device

Fixed extraction of spectra to correctly process the filenames, which include "_nn" for detector number in an array as well as the trailing "_n" for sequence number of the list mode data sequence. This fixes potential loss of data into extracted spectra, which may also affect the *Standards Wizard* processing.

Black periodic table "elements" under Linux

A strange behaviour under Linux leads to some element tiles in periodic elements (*Identify* and *Xray spectrum fit* windows) going black after some hours, like the system's "expose" updates get lazy and stop arriving. A fix forces these to be done, which should fix this Linux only behaviour.

Version 8.9j

Zoom and Pan on Image

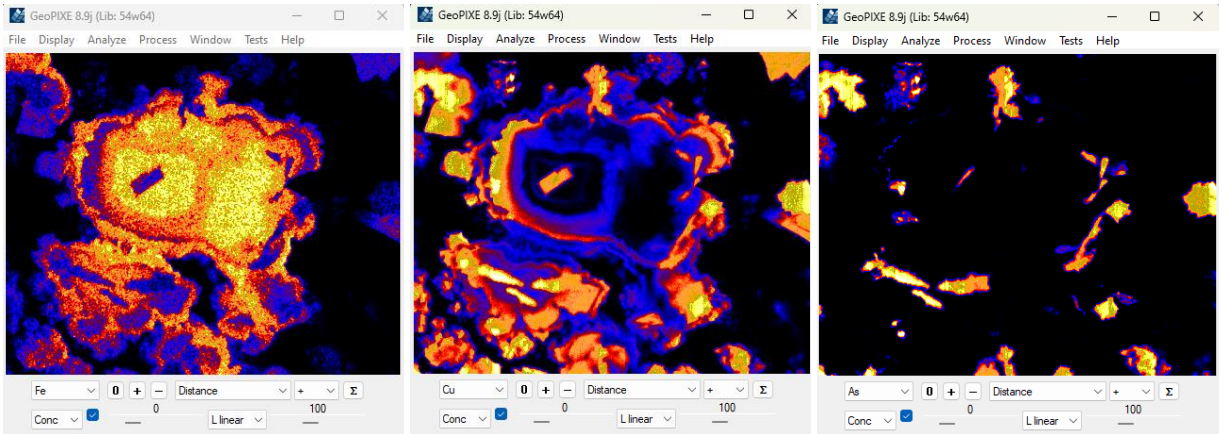
Use ***Control-Left mouse*** to move the image. Control-Left click on a feature and drag to where you'd like that feature to be moved to. Nothing is drawn. On release the image is moved. For example, control-left click and hold on a feature and drag it and release at the centre of the window.

Use ***Control-Scroll wheel*** to increment the Zoom level in and out. The display zooms around the current centre.

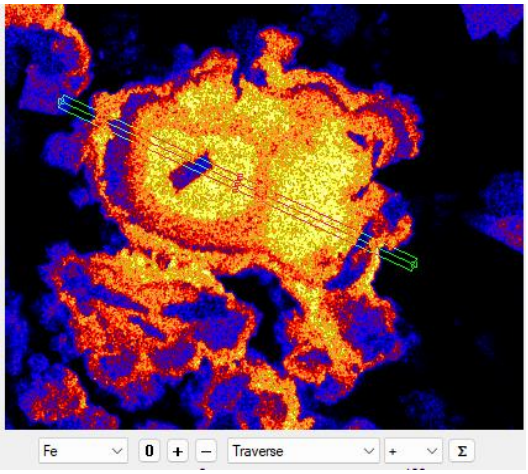
Refine pixel selections for a selective Traverse

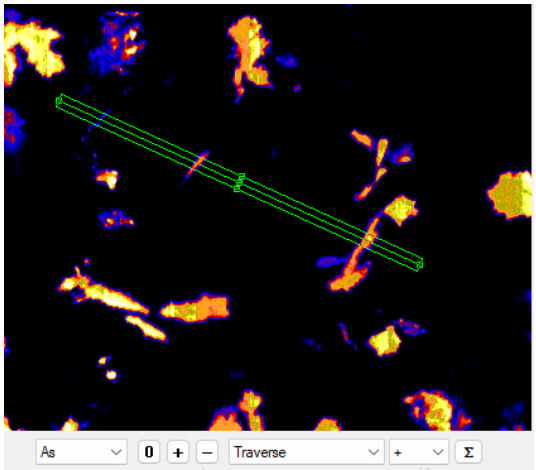
If you would like to look at the variation of a particular element constrained to just a certain mineral phase, for example, ignoring other phases along a *Traverse* line, then this is for you.

Look at the worked example dataset (file "Demo/CSIRO/c4-x/c4-x-3-b.dai"), which has a solid-solution series in Fe versus Cu sulphides varying from near pyrite (FeS_2) through to near pure villamaninite (CuS_2), as well as some enargite (Cu_3AsS_4).

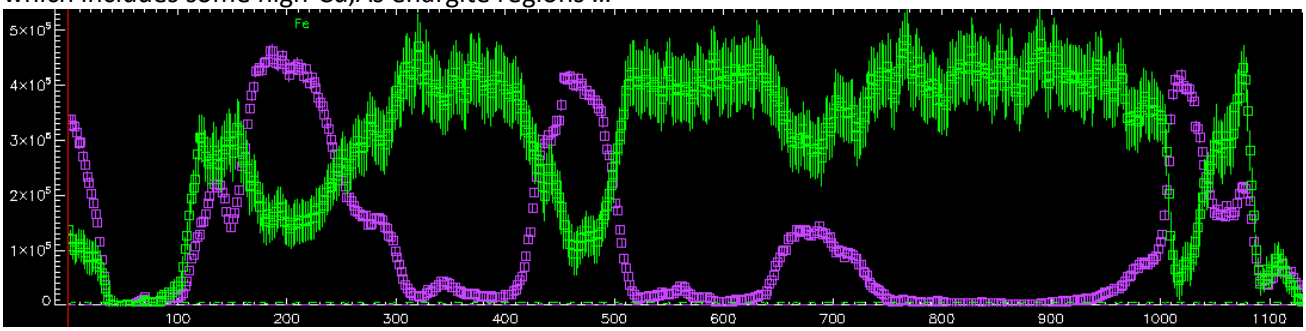


We want to look at the variation of Cu and Fe across the solid-solution members and ignore the enargite. A normal traverse looks like this …



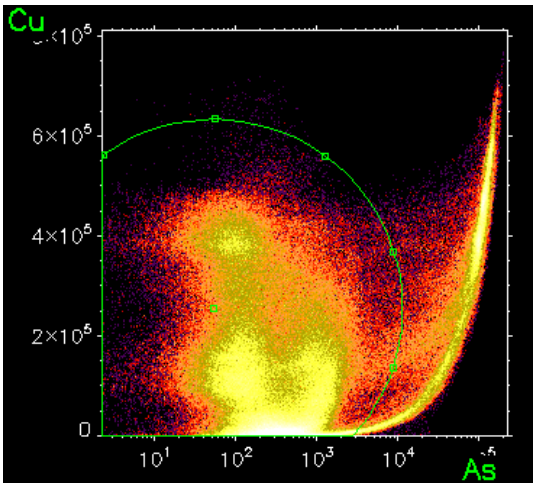


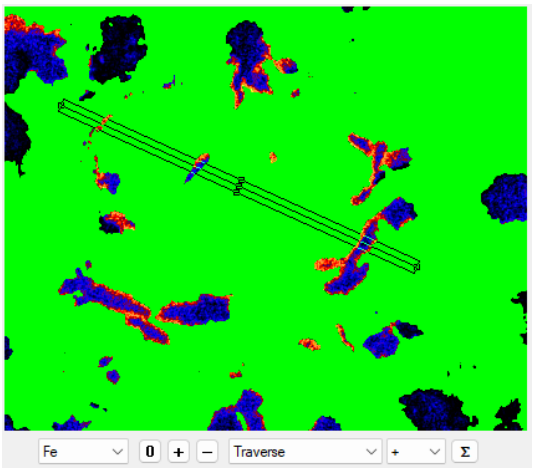
which includes some high-Cu,As enargite regions …



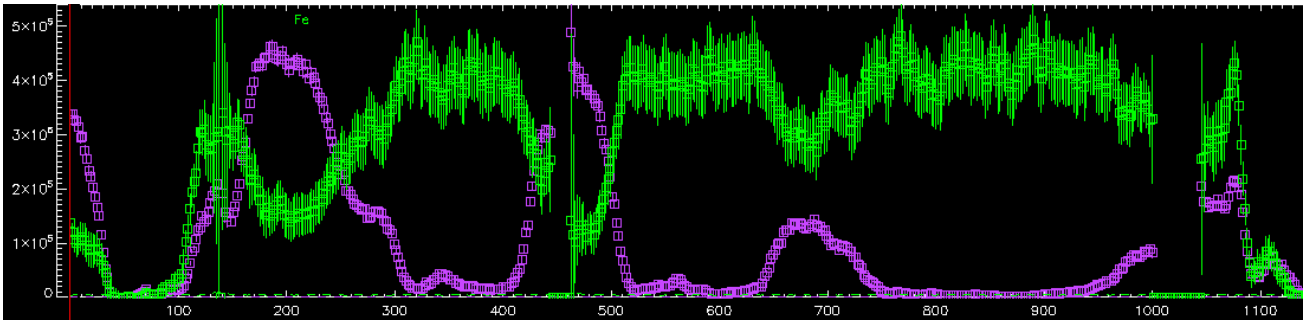
Fe (green), Cu (violet)

Constraining our pixels to exclude the enargite, using an *Association* window plot of Cu versus As, we can avoid the enargite areas …





Now you can show a *Traverse* that skips those regions by using the new Image window menu "**Analyze->Further refine highlighted pixels within spline**", which just refines the green *Association* selected pixels constrained within the *Traverse* rectangle …



Fe (green), Cu (violet)

Version 8.9g

New Help windows

The "Help" menus on the *Image* and *Spectrum Display* windows have been updated to open embedded browser windows showing aspects of the GeoPIXE documentation, such as *GeoPIXE Overview*, the *GeoPIXE Manual*, *Maia detector* notes and the *GeoPIXE Worked Examples* and including the open source GeoPIXE *GitHub* link and links to the *Worked Examples data* on CSIRO DAP.

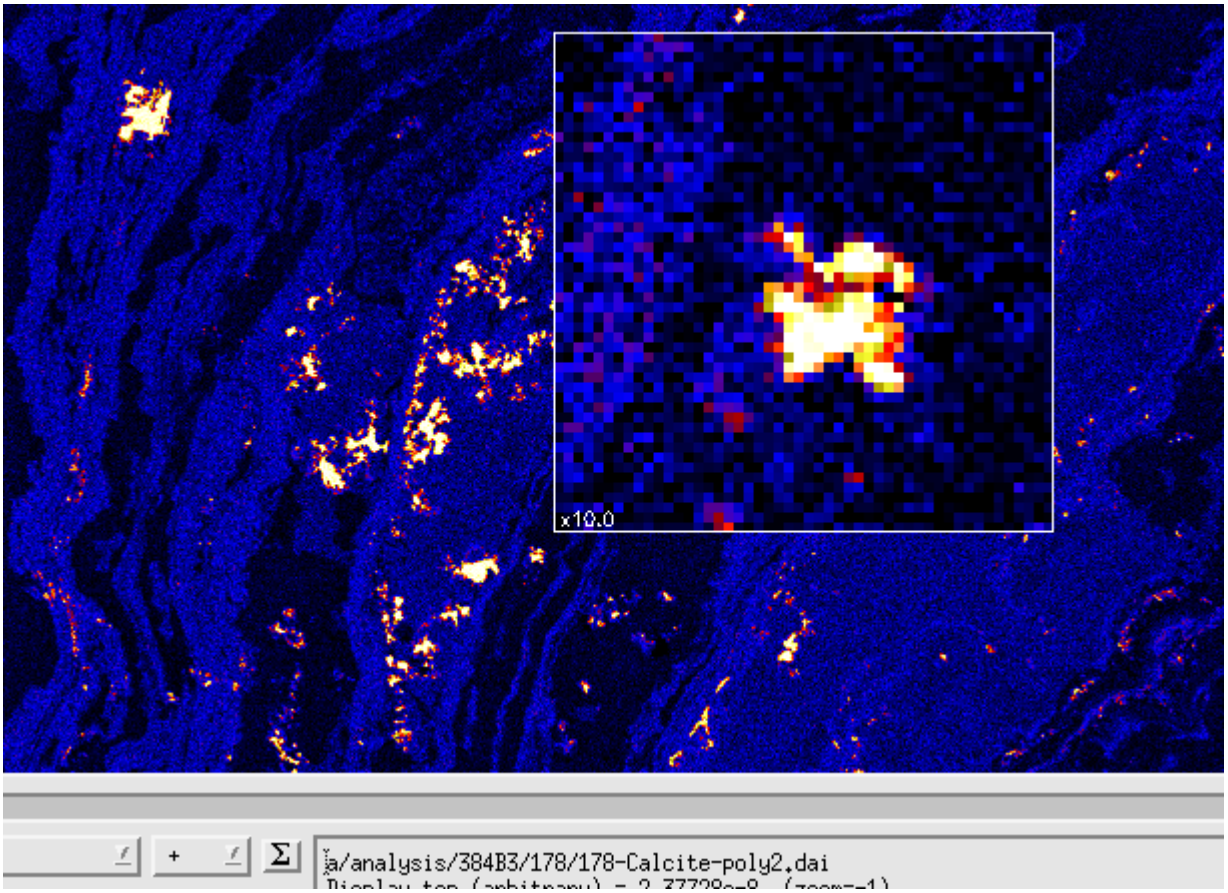
Window specific help

Small "?" buttons in the bottom-right corner of most windows will launch the new help browser for the *GeoPIXE Users Guide* and jump to the section for that window.

Version 8.9e

Magnify Zoom in the Image window

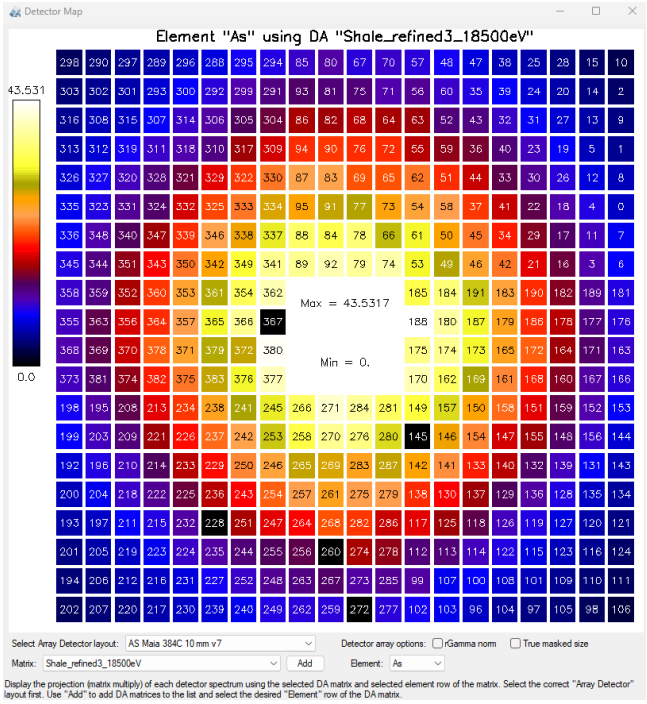
Holding down the *Middle Mouse* button (or zoom *scroll wheel*) will show a magnified view of the pixels near the cursor in the *Image* window. The zoom will always show magnified pixels regardless of the current 'zoom' level. Click and drag to move the "magnifier" across the image.

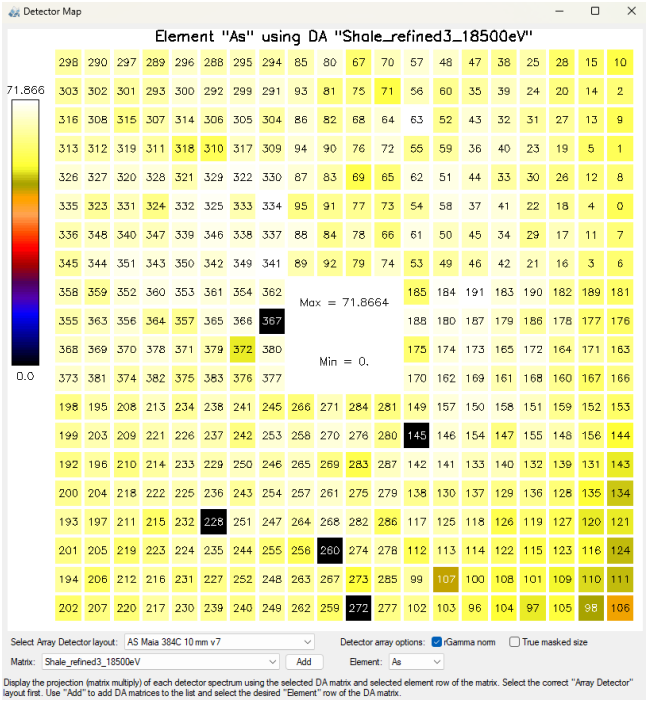


Version 8.9d

Detector Map

A new *Detector Map* window (see “Detector Map” on the “Windows” menu on the *Spectrum Display* window) can be used to show the count-rate (or element concentrations) variation across the Maia detector array (or any array detector). It processes spectral loaded into *Spectrum Display* for individual detector channels. You can load and select a DA matrix file, which is used to transform these spectra into element signals, detector by detector, using a selected array detector layout spec. The model variation in *Relative Sensitivity* across the array (rGamma) can be normalized out with the “rGamma norm” option.





Colour Maps

Added colour tables for the “Inferno” and “Viridis” colour maps.

X,Y Histograms in Association

Added X and Y projection histograms to the *Association* window. Access using the tabs on the right side.

Batch Wizard

A new Wizard has been added as a more extendable replacement to *Batch Sort*.